Discretionary Equity Investing

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Who are the Discretionary Equity Investors?

- Discretionary equity investors use **fundamental analysis** at least since the work of Graham and Dodd ("Security Analysis", 1934)

- Hedge funds doing discretionary investing belong to the **Long-Short Equity** category

  - Typically more long than short. It is **easier to take long positions**. Even if there is no alpha, you can still **earn the equity premium**

- **Dedicated Short hedge funds** are also discretionary focusing on 'bad' companies

- **Long-only investors**, i.e. active mutual funds, can overweight and underweight companies relative to the index

Steps in the Investment Process

1. **Screen the universe** of securities using some criteria, e.g. **valuation ratios** such as P/E ratio or Intrinsic Value-to-Price ratio (as in the Valuation 50 fund's case study in the Appendix)

2. Make a **fundamental analysis** of the selected securities also using equity valuation models (see later)

   - Relies on both **hard information** (data) and **soft information** (conversations with analysts and stakeholders)

3. Make **final investment decision** in analyzed securities based on an the analyst's overall assessment

   - You see that the process being highly discretionary **cannot be backtested**

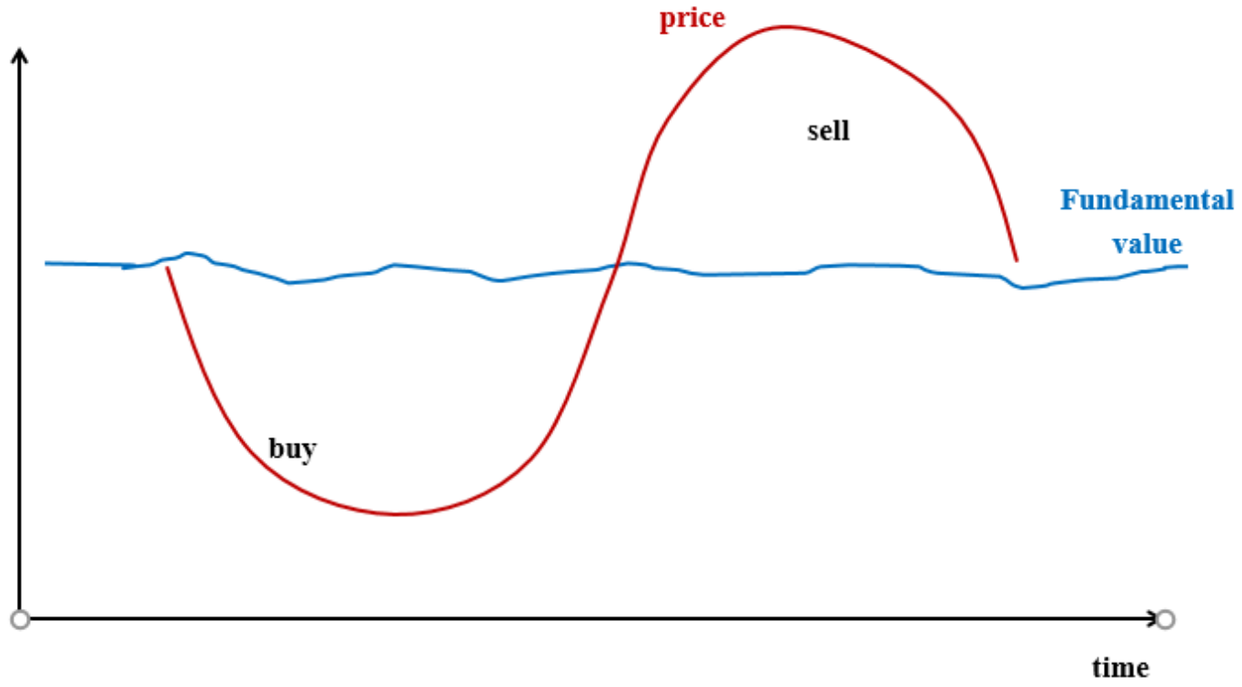
Types of Discretionary Strategies

- **Value Investing**
- **Quality Investing**
- **Search for catalysts**
- **Activist** (done by hedge funds)

Strategy: Value Investing

- Strategy consists of:
  - Buy low (stocks with high intrinsic value/market value)
  - Sell high (stocks with low intrinsic value/market value)
- How to find the intrinsic value:
  - **Hard information**: fundamental analysis using valuation models
  - **Soft information**: talk to the firm and **everyone involved in its "value chain"**
    - Management, employees, unions, customers, suppliers, competitors

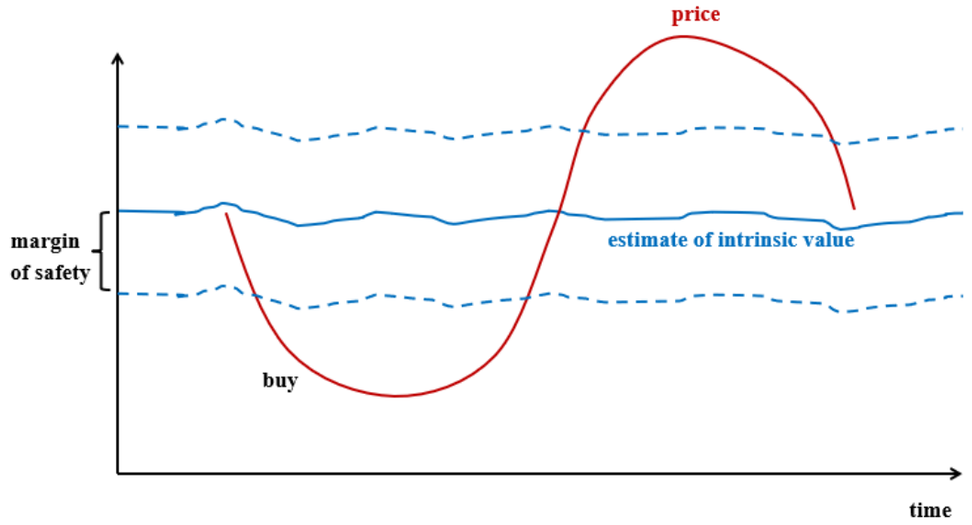
A Value Investor's Perspective



Margin of Safety

- The estimates of intrinsic value can be subject to error. The inputs (cash flows, discount rates) can be wrong. The model assumptions can be wrong

- To be protected from these errors, the recommendation is to adopt a **Margin of Safety**



The Value Trap

- A stock can **look cheap** because
  - It is **undervalued** by the market
  - Or, because its **fundamentals will collapse soon** and the market is anticipating this fact?
- For instance, suppose a stock has a very **low P/E**
- What will adjust to re-establish a normal P/E
  - **P↑ or E↓?**

- In the second case, i.e., $\mathbf{E}\downarrow$,we talk about a **Value Trap**

- A Value Trap is a firm that has low valuation and **will never recover**

Strategy: Quality Investing

- The Value Trap problem can be **mitigated** by combining **Value with Quality Investing**

- Recall Gordon's Growth Model:

$$V_t = \frac{E\left(D_{t+1}\right)}{k-g} \tag{1}$$

- Multiply and divide by Net Income (NI) and divide each side by book value B_t:

$$\begin{aligned} \frac{V_t}{B_t} &= \frac{\frac{E(NI_{t+1})}{B_t} \times \frac{E(D_{t+1})}{E(NI_{t+1})}}{k-g} \\ &= \frac{\text{profitability} \times \text{payout ratio}}{\text{required return - growth}} \end{aligned}$$

- The formula identifies the **4 main "quality" characteristics**, i.e., characteristics **that justify a higher valuation ratio**:

  1. **Profitability** $(\frac{NI}{B})$
  2. **Growing profits** (g)
  3. **Safety and stable profits**, i.e. low required return (k)
  4. Profits are paid to shareholders (Payout Ratio $\frac{D}{NI}$). That is, the **governance is good**

- Let us analyze each of these four characteristics

- Quality investors look for **sustainable profits**, which are not just the result of accounting assumptions and one-off events

- **Accounting assumptions** determine the quality of profits. E.g., high accruals are a sign of bad quality
  - **Accruals** capture the **difference between Earnings and Cash Flows**. Positive accruals result from anticipating the recognition of future revenues and postponing the recognition of current costs
  - E.g. High accruals can result from failing to recognize salary bonuses earned this year but paid out next year

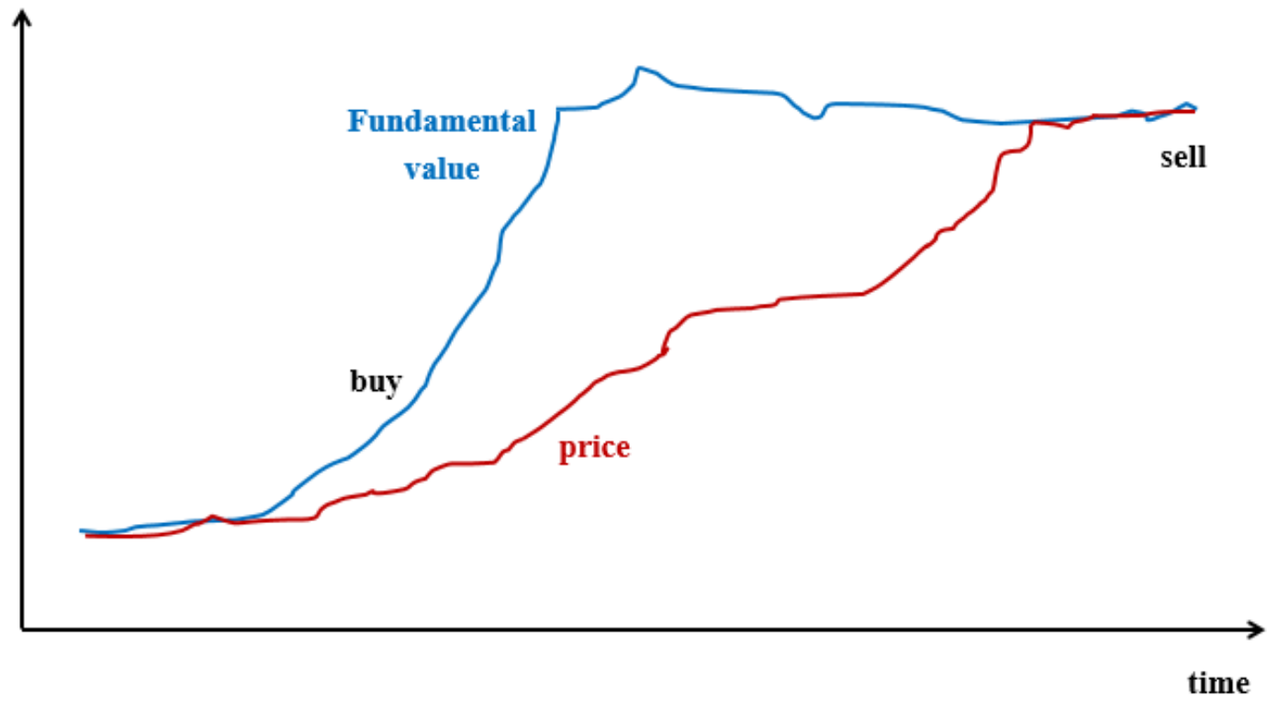
2. Growth at a Reasonable Price (GARP)

- Growth **may be a desirable** characteristic
- An important issue is **whether the price already reflects g or not**
- Growth at a reasonable price (**GARP**) **strategies** search for companies whose prices do not reflect the growth yet
- You start by looking for firms with **good growth**:
  - Consistent past growth
  - Excellent future growth prospects

- Then, you **perform a valuation** to see if the price already reflects this growth or, on the contrary, the stock appears cheap relative to its potential growth

- Avoid **bad growth**:
  - **Value destroying: return on investment $<$ cost of capital**
  - **Empire building**: acquisitions that only benefit the managers who oversee a bigger firm
  - Sales growth with **heavy price discounts**: deterioration of profit margins

A Growth Investor's Perspective



3. Safety

- **Lower discount rate** for safer stocks
- Safety is measured from:
  - **Returns**
  - **Fundamentals**
- Safer firms are those with less volatile fundamentals, **lower financial and operating leverage**
  - Financial leverage comes from **borrowing money**
  - Operating leverage originates from **large fixed costs**. E.g., workforce costs

4. Payout and Management Quality

- Assess whether **profits are paid out** to shareholders through dividends and repurchases

- Or, does the management extract **private benefits**?
  - **Empire building**: wasteful acquisitions
  - **Petty expenses**: corporate jets, new offices, etc.

- **Quality of governance**
  - Independent directors vs. cronies of CEO

  - Entrenched management vs. contendible firm

- **Quality of management**
  - Vision
  - Motivate employees
  - Cut costs
  - Sustainable growth

- These considerations involve **discretionary judgement** that is not part of a quantitative strategy. Therefore, discretionary strategies **cannot be backtested**

Strategy: Search for Catalysts

- Some investors buy a cheap stock with the idea of **holding it for the long term**
  - E.g., **Warren Buffet**: "*...Our holding period is forever.*"
- Others prefer to buy cheap stocks that are **about to experience price increases**
- They look for a **catalyst**
  - An **event** that will make other **investors realize the stock is undervalued**, so that the price will finally catch up with fundamental value
  - E.g. the next earnings announcement will be positive. Or: the approval of a new drug by the FDA

- Some impatient investors can **create a catalyst**
  - E.g. a short seller may use a private investigator to find out about occupancy rates at a hotel chain and then **publicize the report** on the website Seeking Alpha

Strategy: Activist Investing

- **Buy** shares in **a company that could be worth more with better management**
- Then try to **affect its direction**
  - After buying more than 5% of shares, the activist investor must make 13D filing (in the US) and **declare** whether they intend to be active
  - This can already **send a message** to management and the market
- Intervene with **communications and proxy fights** to:
  - **Get rid of management** and/or change board members

- **Give cash back** to shareholders (dividends or share repurchases)
- **Cut waste**
- **Sell assets** that are worth more elsewhere
- **Put the company up for sale**

Equity Valuation Models

The Intrinsic Value

- The **intrinsic value** is the theoretical value at which the security should trade if it was **correctly priced**

- 'Correctly priced' means that the **expectations of the level and risk of futures cash flows are correct**

- There are **two main approaches** to computing Intrinsic Value. Good analysts **use both**

  1. **Discounting** future payoffs, which can be measured as **Dividends**, or **Free Cash Flows**, or **Economic Profits**

     – Free Cash Flows = Operating Cash Flow – Capital Expenditures

– Economic Profits = Operating Profits – Cost of Capital

2. **Relative valuation models** – i.e., pricing against comparables

Dividend Discount Models

- We denote the intrinsic value by V_t and define it as the discounted value of the infinite stream of future cash flows, using k as discount rate

$$\begin{aligned} V_0 &= \frac{E(D_1)}{1+k} + \frac{E(D_2)}{(1+k)^2} + \cdots + \frac{E(D_t)}{(1+k)^t} + \cdots \\ &= \sum_{t=1}^{\infty} \frac{E(D_t)}{(1+k)^t} \end{aligned} \tag{2}$$

- The challenge in making this formula operational is to predict the infinite stream of cash flows

- We need to make further assumptions

The Gordon Growth Model (GGM)

- This model is also called 'Constant Dividend Growth' model

- The assumption is that dividends grow at the constant rate g forever

$$D_{t+1} = (1+g)\, D_t \tag{3}$$

  where we omit the expectation operator $E(\cdot)$ for notational simplicity, but keep in mind that when we talk about future dividends, the expectations is always assumed

- Replacing equation equation (3) into equation (2), after some algebra, one obtains the intrinsic value according to the Gordon Model

$$V_0 = \frac{D_1}{k-g} \tag{4}$$

Note: you have to use the result $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q}$, if $q < 1$. Hence, to obtain equation (4), we **need to assume that** $k > g$

- We can also express the model in terms of the current dividend D_0

$$V_0 = \frac{D_0 (1+g)}{k-g} \tag{5}$$

- Once you obtain V_0, you compare it to the actual market price for the security P_0

- In case, $V_0 > P_0$ the security is underpriced and it may make sense to overweight it in your portfolio. In the opposite case, the security is overpriced, and it may make sense to underweight it or not hold it

- Obviously, a crucial step in implementing equation (4) is the choice of g and k

The discount rate

- CAPM is the most popular model to compute the expected return on an asset

$$k = R_f + \beta E\left(R_{mkt} - R_f\right)$$

- Multi-factor models have also been developed and should be used for robustness

- In our applications, we trust CAPM as giving us the correct measure of risk

Growth expectations

- g is the growth rate of dividends
- Typically, however, you will find forecasts for the growth rate of **Earnings**
- Moreover, some firms do not pay dividends, at least for a part of their life cycle
- Assuming a constant dividend payout ratio, the growth rate of dividends equals the growth rate of earnings

Application of the Gordon Model with Yahoo Finance data

- Let us focus on **Coca Cola**, a company with stable dividends. The price on October 19, 2026, is **$93.90** (Yahoo Finance, quote page of the ticker KO)

- Yahoo Finance reports a forward annual dividend of $2.12 (quote page, "Forward Dividend & Yield")

- In the Analysis tab of Yahoo Finance (table Growth Estimates, row "Next Year"), the analysts expect EPS (earnings-per-share) to grow by 6.66%. We use it as the growth rate g

- Yahoo Finance reports a beta of 0.32, computed on 5 years of monthly data (quote page, "Beta (5Y Monthly)")

Multi-stage Dividend Discount Model

- Some firms have growth forecasts that are very high, at least in the near term: $g > k$
- This high growth forecast, possibly exceeding k, are not sustainable in the long run
- A firm cannot grow faster than the whole economy indefinitely. Otherwise, the firm would have an infinite valuation
- So, it is more realistic to assume that the firm grows at a high rate for a limited number of years and then the growth rate stabilizes at a normal level

- For the first T_1 years, the growth rate is g_1. From $T_1 + 1$ onwards, the growth rate is g_2

- Under these assumptions, we can compute the intrinsic value at time 0 as the sum of the discounted dividends paid up to T_1 and the discounted intrinsic value at time T_1, which depends on the dividends that are paid from $T_1 + 1$ onwards

$$V_0 = \frac{D_0 (1 + g_1)}{1 + k} + \cdots + \frac{D_0 (1 + g_1)^{T_1}}{(1 + k)^{T_1}} + \frac{V_{T_1}}{(1 + k)^{T_1}}$$

  where V_{T_1} can be computed using the Gordon model as

$$\begin{aligned} V_{T_1} &= \frac{D_{T_1} (1 + g_2)}{k - g_2} \\ &= \frac{D_0 (1 + g_1)^{T_1} (1 + g_2)}{k - g_2} \end{aligned}$$

- $T_1 = 10$ years

- From Yahoo Finance, we use the analysts' growth estimate for next year, 6.66%, as the growth rate g_1 for the next T_1 years
  - Note: Yahoo Finance no longer reports a 3-to-5 year growth estimate; the next-year estimate is the longest horizon available, which makes **ten years at** g_1 a strong assumption

- Also, we assume $g_2 = 4\%$, which is similar to the long-term real GDP growth rate plus inflation

of T_2 years (e.g. 5 years) over which the growth rate declines linearly from g_1 to g_2, as Bloomberg's function **DDM** does

Identifying alpha

- Once you have identified a mispriced security, you would like to have an estimate of its alpha
- The alpha is useful as an input in portfolio formation, e.g. in the Treynor-Black model
- You need to make assumptions on the number of years it will take for the mispricing to be corrected
- The extreme assumptions are that correction will take place:
  - in one year
  - over an infinite number of years

Mispricing corrected in one year

- The security is mispriced at time 0

$$P_0 \neq V_0$$

- One makes the assumption that the next period the mispricing will be entirely corrected

$$P_1 = V_1$$

where, using the Gordon Model, we have

$$V_1 = \frac{D_2}{k-g} = \frac{D_1\,(1+g)}{k-g} \tag{6}$$

- If the price is expected to converge to V_1 in one period, then the expected return over that period is

$$E(R_1) = \frac{V_1 + D_1}{P_0} - 1 \tag{7}$$

- So, the alpha, defined as the expected return in excess of the required return k is

$$\alpha = E(R_1) - k = \frac{V_1 + D_1}{P_0} - 1 - k \tag{8}$$

- Replacing Equation (6) into Equation (8), we have that

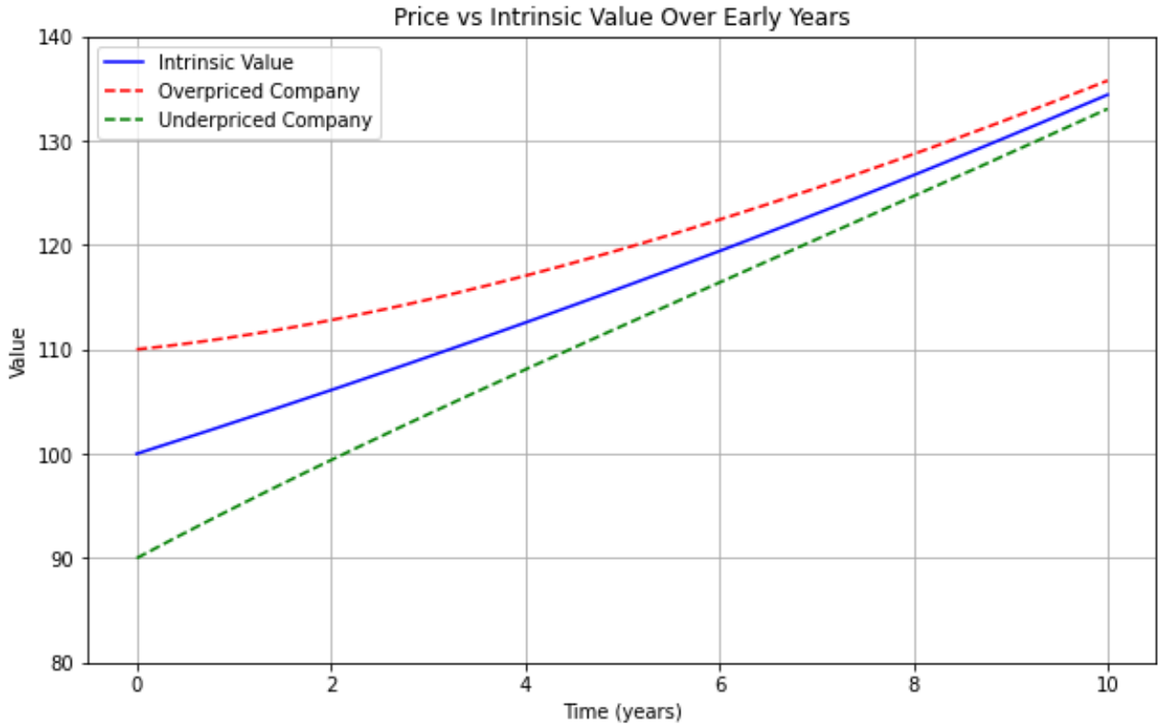
$$\alpha = (1 + k)\frac{V_0 - P_0}{P_0} \tag{9}$$

- Equation (9) holds for any dividend discount model, not only for the Gordon model, because $V_0 = (D_1 + V_1)/(1 + k)$ in all of them. Using the two-stage

Mispricing corrected over infinite horizon

- The mispricing is never entirely corrected and the security has an abnormal return every period

- If the stock is overpriced, it approaches the intrinsic value from above. If it is underpriced, it approaches it from below



- So, every period, the security earns a constant expected return equal to the required return (k) plus an abnormal return (α)

$$E(R) = k + \alpha$$

- Because this expected return is constant, it is the internal rate of return (IRR) that makes the present value of cash flows equal to the current market price

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + E(R))^t}$$

- To simplify, we assume that the dividends grow at the constant rate g

$$\begin{aligned} P_0 &= D_0 \sum_{t=1}^{\infty} \frac{(1+g)^t}{(1+E(R))^t} \\ &= D_0 \frac{1+g}{E(R)-g} \end{aligned}$$

- So, we can solve for $E(R)$

$$E(R) = \frac{D_0(1+g)}{P_0} + g$$

Relative Valuation Models

- You compare a set of firms within an industry to understand which one is over/underpriced relative to the industry average valuation

- To get the price of stock i, use multipliers/ratios – P/E, P/B, P/S – as follows

$$P_i = EPS_i \times P/E_{Comparables}$$

- In the case of Coca Cola, the consensus forecast of EPS for next year is 3.52 (Yahoo Finance, tab Analysis, table Earnings Estimate)

- For the forward P/E of the comparable firms (i.e. the competitors), Yahoo Finance reports it on the Statistics tab of each ticker

Appendix

Appendix: Case Study, Applied Finance Valuation 50

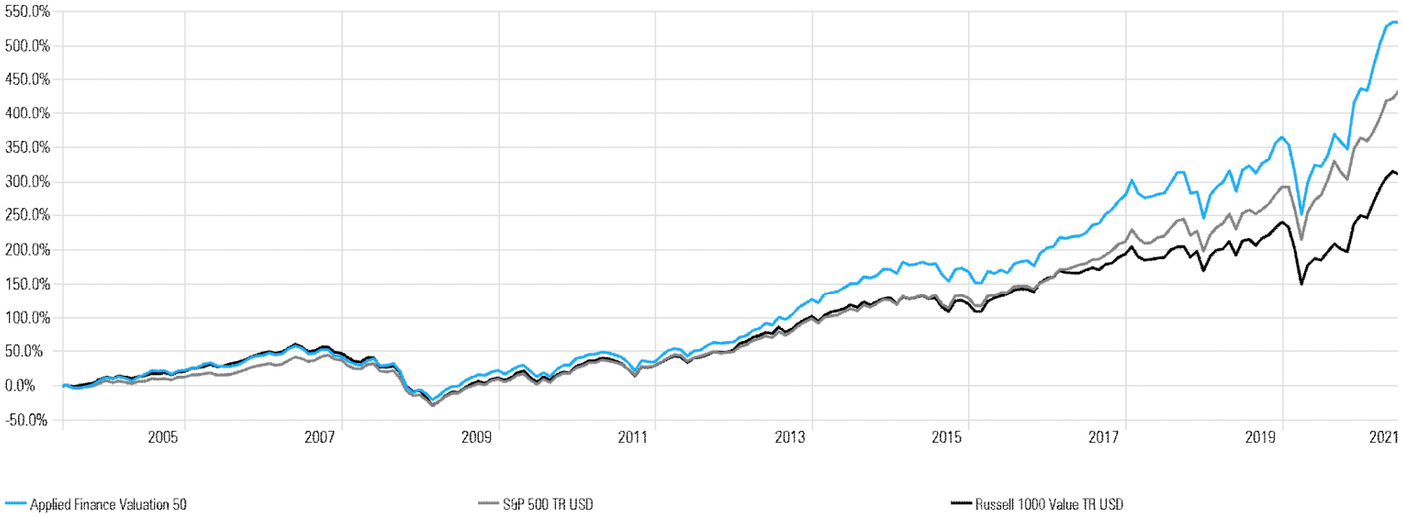
(see full document on website)

- Mutual fund: Always holds 50 stocks to provide effective diversification
- Sector Neutral relative to S&P500
- Equal weighting stocks within sector to focus on owning diversified exposure to attractive valuation, not size
- Quarterly rebalancing to maintain sector and stock weighting discipline
- Aim for discretionary turnover $<=$ 20%, to save transaction cost and maximize tax efficiency

Performance since Inception

Investment Growth

Time Period: 6/11/2004 to 7/31/2021



Trailing Returns

As of Date: 7/31/2021 Data Point: Return Calculation Benchmark: Russell 1000 Growth TR USD

	1 Year	3 Years	5 Years	10 Years	15 Years	Inceptio
Applied Finance Valuation 50	46.81	17.35	18.19	16.37	11.36	11.4
S&P 500 TR USD	36.45	18.16	17.35	15.35	10.86	10.4
Russell 1000 Value TR USD	39.32	11.27	11.41	12.08	7.90	8.6

Steps in Buy Selection Process

APPLIED FINANCE

VALUATION 50

Buy Selection Process

STEP 1 Idea Generation

Baseline valuation screen for top 30% undervalued stocks within each sector

STEP 2 Qualification Analysis

Qualifying buy candidates for best long-term investments

- What is historic Economic Margin level, trend, variability?
- What is historic growth strategy – organic, acquisitive, combination?
- What is management's overall wealth creation track record?
- Do we explain historical valuation well?
- What type of margins, capital structure does the company command?

STEP 3 In-depth analysis

Thoroughly vetting buy candidates that passed qualification analysis

- Understanding business cycle, company evolution, historic strategies
- Conducting industry, peer analysis, assessing company positioning
- Assessing management credibility, competency
- Understanding historic Economic Margin, capital growth record
- Analyzing current business momentum, challenge, and outlook
- Detailed model calibration for intrinsic value estimates, valuation sensitivity analysis.

Buy Selection Process (continued)

APPLIED FINANCE

VALUATION 50

Buy Selection Process

STEP 4 **Decision Deliberations**	› Intrinsic Value Assessment (absolute valuation and relative valuation) › Near / long term catalysts › Industry concentration / diversification › Correlation analysis within portfolio › Risk addition / reduction to the portfolio
STEP 5 **Committee Decision**	› Analyst presents complete BUY analysis/thesis and models to committee for vote

Sell Discipline

Sell Flag	› Baseline valuation screen for existing holdings in the bottom 50% within sector
Sell Alert	› Existing holdings with negative upside based on analysts' custom model › Significant, negative development impairing our confidence forecasting company's future
Sell Analysis	Focus on valuation drill › What expectations are priced in, how different are they from ours › How realistic are market expectations vs. ours
Decision Deliberations	› Valuation attractiveness of sell vs. replacement idea (absolute and relative valuation assessment) › Fundamental strength / weakness of sell vs. replacement idea › Industry concentration / diversification › Risk addition / reduction to the portfolio in the foreseeable economic environment › Return correlation analysis of sell vs replacement idea with the rest of portfolio
Committee Decision	› Analyst presents complete SELL analysis/thesis and models to committee for vote

Replacement case study: Health Care Sector

Ticker	Name	Industry	Valuation	Mkt. Cap
US:MYL	MYLAN NV	Pharmaceuticals	100%	18,868
US:BIIB	BIOGEN INC	Biotechnology	98%	74,725
US:ABC	AMERISOURCEBERGEN CORPORATION	Health Care Distributors	97%	20,002
US:MCK	MCKESSON CORPORATION	Health Care Distributors	95%	26,397
US:CELG	CELGENE CORPORATION	Biotechnology	94%	62,876
US:ABBV	ABBVIE INC	Biotechnology	92%	143,206
US:AGN	ALLERGAN PLC	Pharmaceuticals	91%	64,630
US:LH	LABORATORY CORPORATION OF AMER	Health Care Services	89%	17,715
US:ALXN	ALEXION PHARMACEUTICALS, INC.	Biotechnology	88%	30,971
US:REGN	REGENERON PHARMACEUTICALS INCO	Biotechnology	86%	43,618
US:MTD	METTLER-TOLEDO INTERNATIONAL I	Life Sciences Tools & Services	84%	15,355
US:ESRX	EXPRESS SCRIPTS HOLDING COMPAN	Health Care Services	83%	53,405
US:HOLX	HOLOGIC, INC.	Health Care Equipment	81%	11,141
US:IQV	IQVIA HOLDINGS INC	Life Sciences Tools & Services	80%	26,272
US:CAH	CARDINAL HEALTH, INC.	Health Care Distributors	78%	16,686
US:HCA	HCA HEALTHCARE INC	Health Care Facilities	77%	48,241
US:HSIC	HENRY SCHEIN, INC.	Health Care Distributors	75%	13,029
US:CVS	CVS HEALTH CORP	Health Care Services	73%	80,058
US:UHS	UNIVERSAL HEALTH SERVICES, INC	Health Care Facilities	72%	11,933
US:PRGO	PERRIGO COMPANY PLC	Pharmaceuticals	70%	9,742
US:ZBH	ZIMMER BIOMET HOLDINGS INC	Health Care Equipment	69%	26,728
US:PFE	PFIZER INC.	Pharmaceuticals	67%	258,343
US:EVHC	ENVISION HEALTHCARE CORP	Health Care Services	66%	5,552
US:MRK	MERCK & CO. , INC.	Pharmaceuticals	64%	189,413
US:CI	CIGNA CORPORATION	Managed Health Care	63%	50,646
US:JNJ	JOHNSON & JOHNSON	Pharmaceuticals	61%	370,614
US:GILD	GILEAD SCIENCES, INC.	Biotechnology	59%	100,064
US:BMY	BRISTOL-MYERS SQUIBB COMPANY	Pharmaceuticals	58%	101,315
US:AMGN	AMGEN INC.	Biotechnology	56%	134,531
US:MDT	MEDTRONIC PLC	Health Care Equipment	55%	132,970
US:DHR	DANAHER CORPORATION	Health Care Equipment	53%	76,029
US:SYK	STRYKER CORPORATION	Health Care Equipment	52%	66,452
US:TMO	THERMO FISHER SCIENTIFIC INC.	Life Sciences Tools & Services	50%	98,315
US:DGX	QUEST DIAGNOSTICS INCORPORATED	Health Care Services	48%	14,676
US:CERN	CERNER CORPORATION	Health Care Technology	47%	21,189
US:BSX	BOSTON SCIENTIFIC CORPORATION	Health Care Equipment	45%	53,180

Buy Candidates Based on Default Valuation Rank

- Sales/earnings decline since 2015
- In decline mode in the foreseeable future
- Valuation uncertainty
- One trick pony
- Big management shake up
- Big strategic uncertainty

Valuation Sell Flag

Replacement case study: Buy Candidates

Ticker	Name	Industry	Valuation	Mkt. Cap
US:MYL	MYLAN NV	Pharmaceuticals	100%	18,868
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US:PFE	PFIZER INC.	Pharmaceuticals	67%	258,343
US:EVHC	ENVISION HEALTHCARE CORP	Health Care Services	66%	5,552

→ Generics, unfavorable industry

→ Recent holding that is dropped

→ Big exposure to HCV, similar to GILD

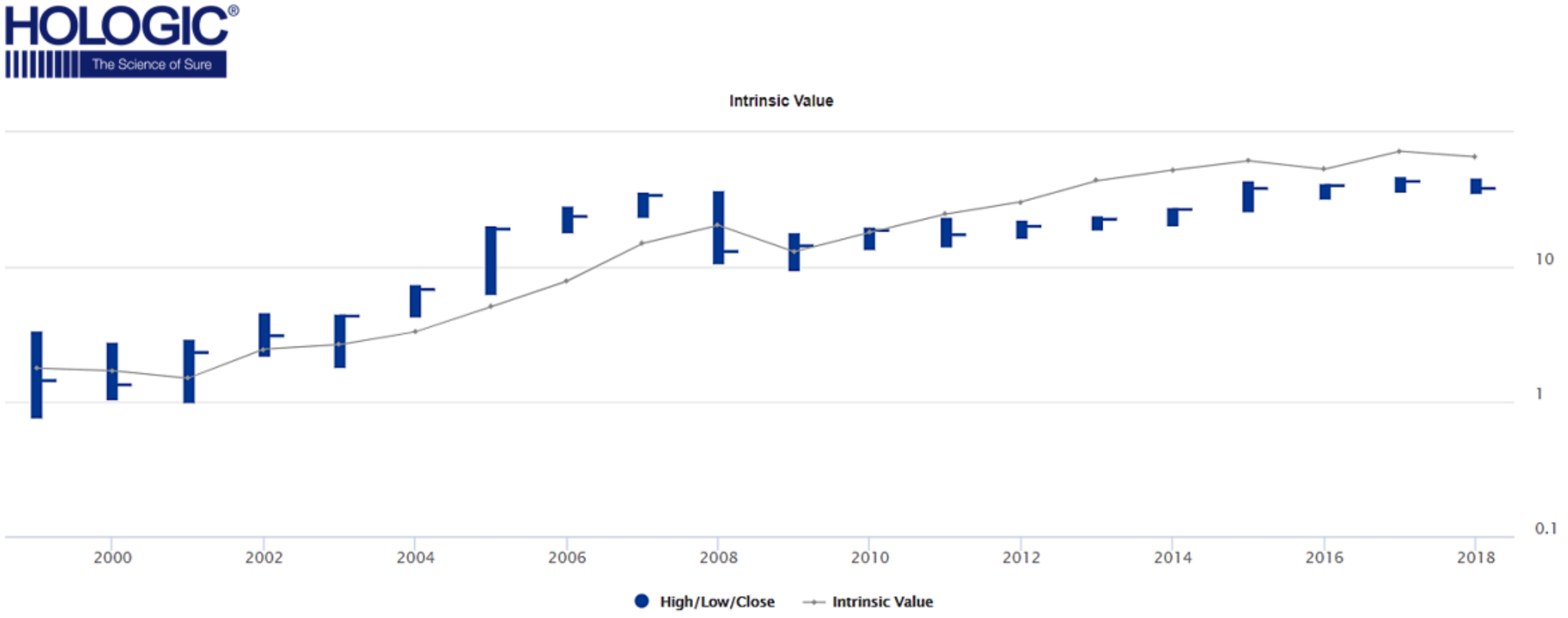
→ To merge with Cigna, similar to CVS

→ Weaker competitor to MCK

→ Unfavorable industry

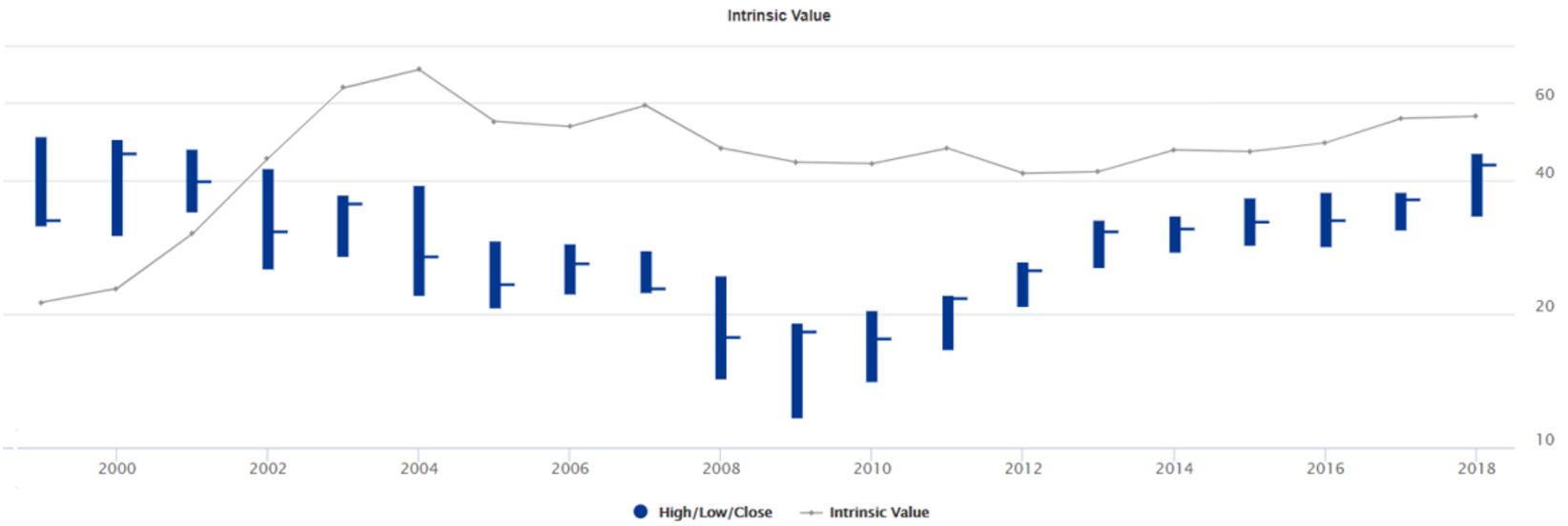
→ Previous holding in unfavorable industry

Replacement case study: Hologic Inc.



- The stock is underpriced. However...

Replacement case study: Pfizer



Appendix: Warren Buffett's Alpha



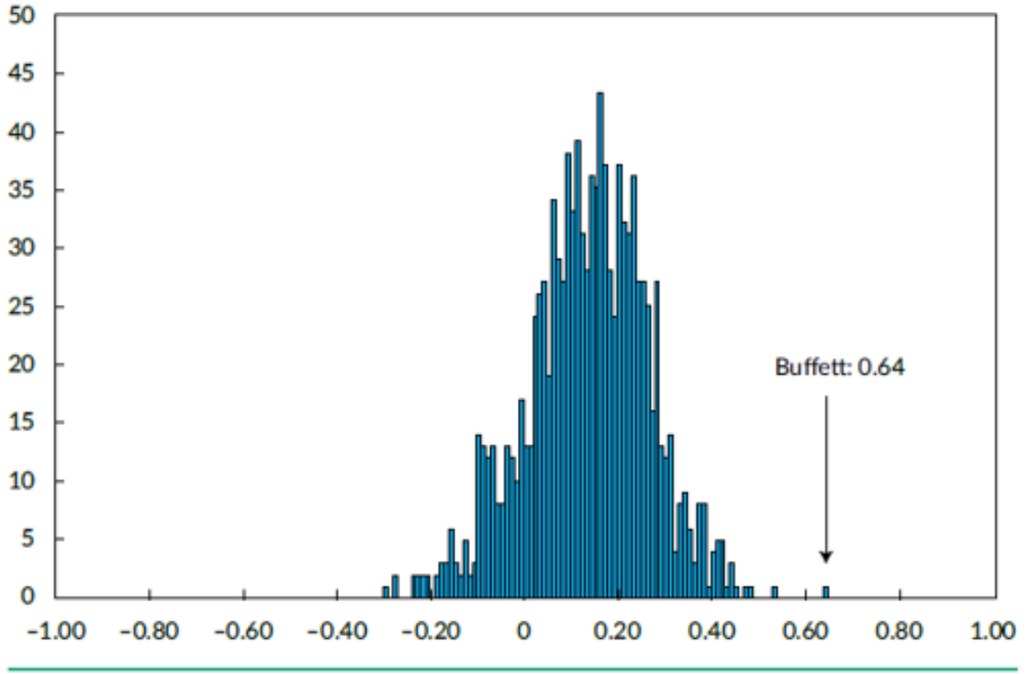
- Using Berkshire Hathaway stock returns and financial filings

- Track record: How good is Buffett?

- Buffett's Leverage?
- Decomposing Buffett: CEO of private companies vs. stock picker?
- Exposures: What type of companies?
- Based on: Frazzini, Kabiller, Pedersen (2013)

How good is Buffett's record among stocks?

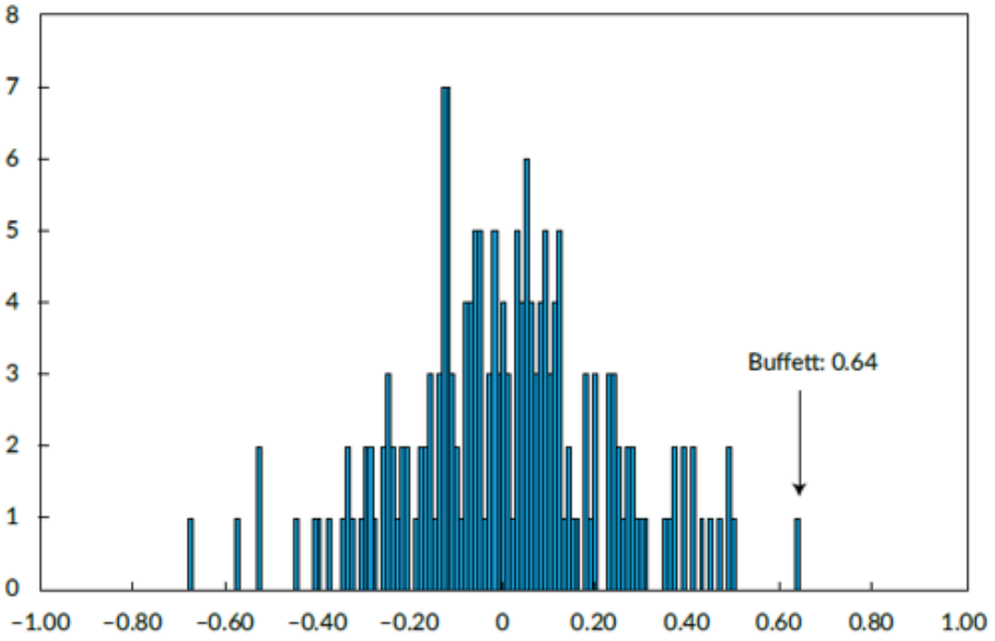
Information ratio of all U.S. stocks with more than 40 years of history: top 1%



Notes: This figure shows the distribution of annualized information ratios of all common stock in the CRSP database with at least 40 years of return history. See also definitions in the notes to Table 1.

How good is Buffett's record among mutual funds?

Information ratios of all U.S. active mutual funds with more than 40 years of history: top 1%



Notes: This figure shows the distribution of annualized information ratios of all actively managed equity funds in the CRSP mutual fund database with at least 40 years of return history. See also definitions in the notes to Table 1.

Buffett's leverage: Magnitude

"I've seen more people fail because of liquor and leverage...You really don't need leverage in this world much. If you're smart, you're going to make a lot of money without borrowing." — Warren Buffett

- Is it really true for Berkshire Hathaway?

- Stylized balance sheet of Berkshire Hathaway:

Assets		**Liabilities**	
Public/private firms	\$380 B	Debt	\$ 63 B
Cash	\$ 47 B	Insurance float and rel.	\$ 85 B
		Deferred tax	\$ 44 B
		Accounts payable	\$ 35 B
		Derivatives sold	\$ 8 B
		Shareholders equity	\$192B
Total	**\$427 B**	Total	\$427 B

- Leverage: L_t=(Total Assets -Cash)/Equity=1.7

- Does Buffett's risk reflect the use of leverage?
  - Volatility of Berkshire Hathaway (1976-2017) : 23.5%
  - Volatility of the portfolio of Berkshire's public stocks (1980-2017):16.2%

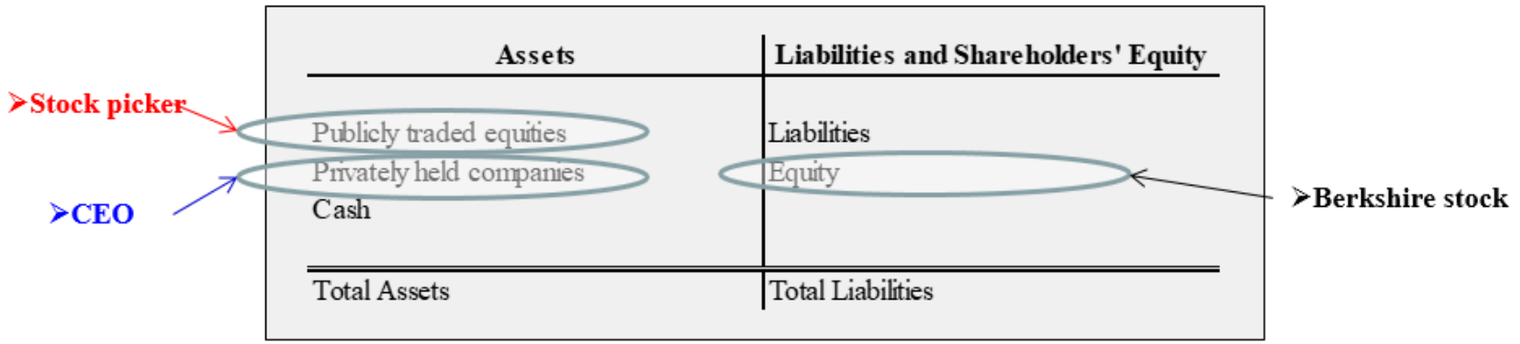
- Hence, leverage does increase the risk of the company relative to the assets, by about 45%

Buffett's leverage: The Source of Leverage

In 2012:

- Debt: rated AAA, 1989-2009, now at AA-
- Deferred income taxes
- Insurance float (36% of liabilities on average): because BH collects insurance premiums upfront and (maybe) pays claims later
  - This is (almost) free. It actually has negative cost if premiums>claims
- Derivatives: Selling options

Decomposing Buffett: CEO of private companies or Stockpicker?



- Returns of:
  - Berkshire stock: observed directly
  - Publicly traded equities: observed via 13F filings and stock return data

– Privately held companies inferred from accounting identity:

$$r_{t+1}^{private} private_t^{MV} + r_{t+1}^{public} public_t^{MV} + r_{t+1}^{f} cash_t = r_{t+1}^{equity} equity_t^{MV} + r_{t+1}^{f} liabilities_t^{MV} \tag{11}$$

hence:

$$r_{t+1}^{private} = \frac{r_{t+1}^{equity} equity_t^{MV} + r_{t+1}^{f} liabilities_t^{MV} - r_{t+1}^{public} public_t^{MV} - r_{t+1}^{f} cash_t}{private_t^{MV}} \tag{12}$$

Better as a Stockpicker than CEO

	Performance			
	Berkshire Hathaway	Public US Stocks (from 13F filings)	Private Holdings	Overall Stock Market Performance
Sample	1976–2017	1980–2017	1984–2017	1976–2017
Beta	0.69	0.77	0.29	1.00
Average excess return	18.6%	12.0%	9.3%	7.5%
Total volatility	23.5%	16.2%	20.6%	15.3%
Idiosyncratic volatility	21.1%	11.2%	20.1%	0.0%
Sharpe ratio	0.79	0.74	0.45	0.49
Information ratio	0.64	0.51	0.35	0.00
Leverage	1.71	1.00	1.00	1.00
Subperiod excess returns:				
1976–1980	41.2%	31.5%		7.3%
1981–1985	28.6	21.3	18.4%	4.8
1986–1990	17.3	12.6	9.6	5.7
1991–1995	29.7	18.8	22.9	12.6
1996–2000	14.9	12.1	8.7	12.1
2001–2005	3.2	2.2	1.8	0.9
2006–2010	6.1	4.1	4.0	2.5
2011–2015	10.8	9.9	5.0	12.1
2016–2017	19.3	13.6	11.1	15.0

What kind of companies does Buffett own?

Buffett: "Buy companies with strong histories of profitability" and "consistent operating history"

- Run regression

  $$r_t = \alpha + \beta_1 MKT_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 BAB_t + \beta_6 QMJ_t + \varepsilon_t$$

  where:

  - UMD is momentum factor
  - BAB is betting-against-beta factor, a low-volatility factor
  - QMJ is quality-minus-junk factor